IE1005 – FROM COMPUTATIONAL THINKING TO PROGRAMMING
AY26/27 SEMESTER 1 HOME ASSIGNMENT 2

Instructions

- Save your source code as `HW2_YourName.c` and submit it on your NTULearn tutorial group site.
- You are required to submit a video recording of yourself explaining your source code. Refer to the video recording guide available on NTULearn.
- You may resubmit as many times as needed before the deadline. Only the last submission before the deadline will be graded.
- Ensure you submit the correct files. Your grade will be based on the files you submit.

Question

Write a C program that simulates an embedded thermal management controller for a dual-core CPU (Core 0 and Core 1). Its primary objective is to monitor core temperatures under varying computational workloads, predict thermal runaway conditions, and execute protective system decisions, such as thermal throttling and emergency shutdown, to prevent hardware damage.

The simulation executes over a discrete number of clock cycles. Both cores start at an ambient temperature of 30 °C. In each cycle, user-defined integer Workloads ranging from 0 to 100 are applied to both cores. Active cooling is modeled via a user-selected Fan Tier, 1, 2, or 3.

The following temperature rules apply to each core in each clock cycle:

- Temperature change due to base heat generation, ΔT_{base}
  - If Workload < 30: $\Delta T_{base} = +2$ °C
  - If 30 ≤ Workload ≤ 70: $\Delta T_{base} = +0.15$°C × Workload
  - If Workload > 70: $\Delta T_{base} = +0.25$°C × Workload
- Net-temperature change accounting for the thermal runaway effect, ΔT_{net}
  - If current temperature ≥ 85 °C: $\Delta T_{net} = 1.2 \times \Delta T_{base}$
  - Otherwise: $\Delta T_{net} = \Delta T_{base}$
- Temperature change due to active cooling, ΔT_{cool}
  - If Fan Tier 1: $\Delta T_{cool} = -4$ °C
  - If Fan Tier 2: $\Delta T_{cool} = -9$ °C
  - If Fan Tier 3: $\Delta T_{cool} = -16$ °C
- New temperature
  - New temperature = current temperature + ΔT_{net} + ΔT_{cool}
  - Temperature cannot fall below the ambient temperature floor of 30 °C. If the calculation drops below 30 °C, clamp the value to 30 °C.

The program should do the following. Example user inputs are highlighted in **boldface**.

1. Display a starting message and prompt the user to enter the total simulation clock cycles and the cooling Fan Tier.

```
=== CPU Thermal Controller Simulation ===

Enter number of cycles: 5
Enter Fan Tier (1-3): 2
```

2. In each clock cycle, prompt the user for core Workloads. Compute and display the updated core temperatures formatted to 2 decimal places.

```
--- Cycle 1 ---
Enter workloads for Core 0 and Core 1 (0-100): 40 80
```

```
Core 0: 30.00 | Core 1: 41.00
```

3. At the end of each clock cycle, evaluate temperatures for safety violations, and take the following actions as necessary:
   a. If either core reaches or exceeds 85 °C, trigger a throttling alert.

```
⋮
Core 0: 75.75 | Core 1: 86.75
[ALERT] Thermal Throttling Active!
```

   b. If both cores exceed 95 °C simultaneously, immediately shutdown the system.

```
⋮
Core 0: 97.05 | Core 1: 104.30
[CRITICAL] Emergency Shutdown!
```

4. Upon cycle completion or after an emergency shutdown, display a diagnostic summary.

```
⋮
Core 0: 85.50 | Core 1: 88.91
[ALERT] Thermal Throttling Active!

=== Diagnostic Summary ===
System Status      : Safe Completion
Total Cycles Run   : 5
Throttled Cycles   : 2
Final Temperatures : Core 0: 85.50 | Core 1: 88.91
```

Below are two examples of the entire program run.

```
=== CPU Thermal Controller Simulation ===

Enter number of cycles: 3
Enter Fan Tier (1-3): 3

--- Cycle 1 ---
Enter workloads for Core 0 and Core 1 (0-100): 80 20
Core 0: 34.00 | Core 1: 30.00

--- Cycle 2 ---
Enter workloads for Core 0 and Core 1 (0-100): 98 92
Core 0: 42.50 | Core 1: 37.00

--- Cycle 3 ---
Enter workloads for Core 0 and Core 1 (0-100): 100 98
Core 0: 51.50 | Core 1: 45.50

=== Diagnostic Summary ===
System Status      : Safe Completion
Total Cycles Run   : 3
Throttled Cycles   : 0
Final Temperatures : Core 0: 51.50 | Core 1: 45.50
```

```
=== CPU Thermal Controller Simulation ===

Enter number of cycles: 5
Enter Fan Tier (1-3): 1
```

```
--- Cycle 1 ---
Enter workloads for Core 0 and Core 1 (0-100): 99 98
Core 0: 50.75 | Core 1: 50.50

--- Cycle 2 ---
Enter workloads for Core 0 and Core 1 (0-100): 100 99
Core 0: 71.75 | Core 1: 71.25

--- Cycle 3 ---
Enter workloads for Core 0 and Core 1 (0-100): 100 100
Core 0: 92.75 | Core 1: 92.25
[ALERT] Thermal Throttling Active!

--- Cycle 4 ---
Enter workloads for Core 0 and Core 1 (0-100): 98 95
Core 0: 118.15 | Core 1: 116.75
[CRITICAL] Emergency Shutdown!

=== Diagnostic Summary ===
System Status      : Emergency Shutdown (Cycle 4)
Total Cycles Run   : 4
Throttled Cycles   : 1
Final Temperatures : Core 0: 118.15 | Core 1: 116.75
```

Note

- Implement a custom function that computes the new temperature of an individual core in each clock cycle. Do not use more than one custom function.
- Do not use global variables.
- Assume that the user always enters valid input.
- Optimize your program for efficiency and readability.
- Remember to style your code appropriately.

Marking schedule

• Prompt the user and handle input successfully	5 marks
• Appropriate use and implementation of the custom function	15 marks
• Correct execution of program logic and control sequence	15 marks
• Program efficiency and readability	10 marks
• Adherence to good coding style (indentation, headers, variable names, etc.)	5 marks
• Recorded presentation and demonstration under 5 minutes	50 marks
Total	100 marks
• Use of global variables	-30 marks

END